

PULMONARY FIBROSIS PROGNOSIS

PROJECT REPORT

submitted by

ABHISHEK T (TVE17CS004)

SAURAV S B (TVE17CS057)

SREERAG M S (TVE17CS059)

HISHAM ALI P (LTVE17CS066)

to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Technology

in

Computer Science and Engineering



Department of Computer Science and Engineering

College of Engineering Trivandrum

Kerala

June 23, 2021

DECLARATION

We undersigned hereby declare that the project report **Pulmonary Fibrosis Prognosis**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Asst. Prof. Thara R J**. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Thiruvananthapuram

Date: June 23, 2021

Abhishek T

Saurav S B

Sreerag M S

Hisham Ali P

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING TRIVANDRUM**



CERTIFICATE

This is to certify that the report entitled "**PULMONARY FIBROSIS PROGNOSIS**", submitted by **Abhishek T, Saurav S B, Sreerag M S, Hisham Ali P** to the **APJ Abdul Kalam Technological University** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Thara R J

Assistant Professor
Department of CSE
(Guide)

Deepthi V R

Assistant Professor
Department of CSE
(Coordinator)

Dr. Sumesh Divakaran

Professor
Department of CSE
(Head of Department)

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Abhishek T

Saurav S B

Sreerag M S

Hisham Ali P

ABSTRACT

Idiopathic pulmonary fibrosis is a condition in which the human lungs become scarred over time. Symptoms of pulmonary fibrosis include shortness of breath, a dry cough, feeling tired, weight loss, and nail clubbing. Complications may include pulmonary hypertension, respiratory failure, pneumothorax, and lung cancer. Respiratory failure resulting from disease progression is the most frequent cause of death.

The aim of this project is to create a machine learning model which gives the prognosis for this disease which can help patients to plan treatment in advance. More accurate prediction of survival in IPF would be useful to facilitate counseling patients and their families, to aid communication among providers, and to guide optimal timing of transplantation. The project will provide critical information such as Forced Vital Capacity (FVC) which will help in understanding how the disease reacts.

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ABBREVIATIONS

(List in the alphabetical order)

CALIPER Computer-Aided Lung Informatics for Pathology Evaluation and Ratings

CT Computer Tomography

DICOM Digital Imaging and Communications in Medicine

FVC Forced Vital Capacity

GHNC Gaussian Histogram Normalized Correlation

HU Hounsified Unit

IPF Idiopathic Pulmonary Fibrosis

MSE Mean Square Error

Chapter 1

Introduction

Pulmonary Fibrosis (PF) describes a condition in which the lung tissue becomes thickened, stiff, and scarred. The medical terminology used to describe this scar tissue is fibrosis. The alveoli and the blood vessels within the lungs are responsible for delivering oxygen to the body, including the brain, heart, and other organs. As lung tissue becomes scarred and thicker, it is more difficult for the lungs to transfer oxygen into the bloodstream. As a result, the brain, heart, and other organs do not get the oxygen they need to function properly. The estimated prevalence of Pulmonary Fibrosis in the India ranges from 14 - 43 per 100,000 populations. The metric score provides an accurate prognosis by providing weekly FVC scores. Our project aims at predicting the future state of a person and thereby helping doctors to create a treatment action plan.

The remainder of this report is organized as follows. In chapter 2, the summary of various existing methods are described. The objective and design and implementations are provided in the following chapters Chapter 4 discusses the result of the project. Finally, chapter5 concludes the report by stating the advantages of our model and the future research directions.

Chapter 2

Existing Methods and Objectives

This chapter discusses some of the research papers on Pulmonary Fibrosis along with the Objectives of our project.

The paper (1) proposes a convolutional autoencoder deep learning framework which uses unlabelled computed tomography (CT) images to extract important features. The framework which is based on convolutional autoencoder neural network (CANN) uses only a small amount of labelled data for fine-tuning the model. The availability of labelled data is rare because of privacy and security issues. The paper shows that the proposed scheme is superior to other approaches, which effectively solves the intrinsic labor-intensive problem during artificial image labeling.

Different computer-based chest computed tomography (CT) image processing methods are used in pulmonary fibrosis patients. The techniques includes density histogram analysis, mean CT value of whole lungs, density mask technique, and texture classification methods. Most of these techniques provide strong evaluation of lung functions, disease development, and mortality. In clinical practice, each approach has its own merits that can be used. In (2) paper, limits and many potential directions of computer based CT image analysis of Idiopathic Pulmonary Fibrosis (IPF) are discussed.

Paper (3) is a complete medical description of Idiopathic pulmonary fibrosis (IPF). This paper explain all the medical periods of Idiopathic pulmonary fibrosis and the development of disease in each stage which is very important for the prognosis of the disease. Several individual clinical variables have been shown to correlate with survival however, there is no

established method of combining these predictors to accurately determine prognosis or define the stage of disease that is why prognosis is that much important for this disease . All the medical predictors of Idiopathic pulmonary fibrosis have high medical value. There are almost eight different types of predictors are available for this disease all the predictors are well explained in this paper. In this project we use some of the clinical predictors mentioned in the paper like age, sex, smoking status and forced vital capacity(FVC)

2.1 DEEP FEATURE LEARNING FOR MEDICAL IMAGE ANALYSIS WITH CONVOLUTIONAL AUTOENCODER NEURAL NETWORK(1)

This paper proposes an unsupervised learning method called CANN.

- Proposes a convolutional autoencoder deep learning framework to support unsupervised image features learning for lung nodule through unlabeled data.
- Model is fine tuned using labelled data.
- Unlabelled data is fed into the encoder part of the Auto-Encoder and the model will try to recreate the same data using the feature map. The difference between the original image and the reconstructed image is the loss.
- This framework is suitable for medical purposes due to the shortage of labelled dataset.

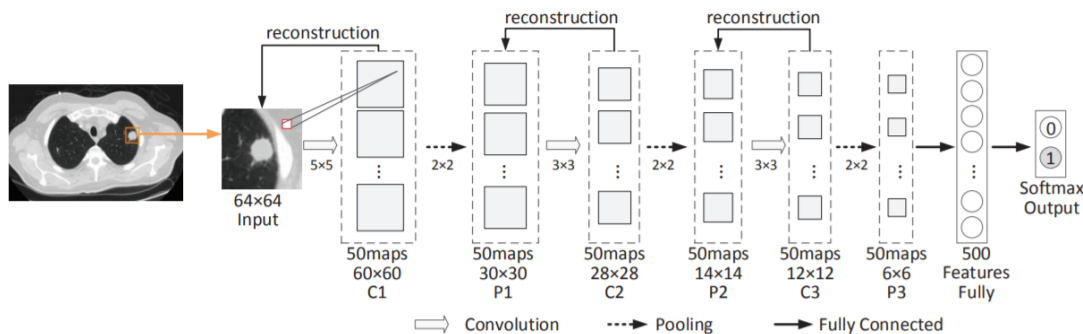


Figure 2.1: Convolutional Autoencoder Neural Network Architecture

The figure 2.1 is the architecture of the convolutional autoencoder. The key points of observation deduced from the given paper (1) is that Auto-Encoder is able to extract latent features from huge dimensions thereby making the process lighter. So it is suitable for building a light and fast model. CANN was also able to deliver 95% accuracy in the classification use case using a small dataset thereby beating every other framework.

2.2 COMPUTER-BASED QUANTITATIVE COMPUTED TOMOGRAPHY IMAGE ANALYSIS IN IDIOPATHIC PULMONARY FIBROSIS(2)

Paper proposes multiple methods of analysing CT images for cases with Idiopathic Pulmonary Fibrosis (IPF). Following are the two major methods of CT Analysis,

2.2.1 The density histogram analysis

In this method volume density histogram of patients lungs. Analyses asymmetry and peakedness in the density histogram of lungs of the patient.

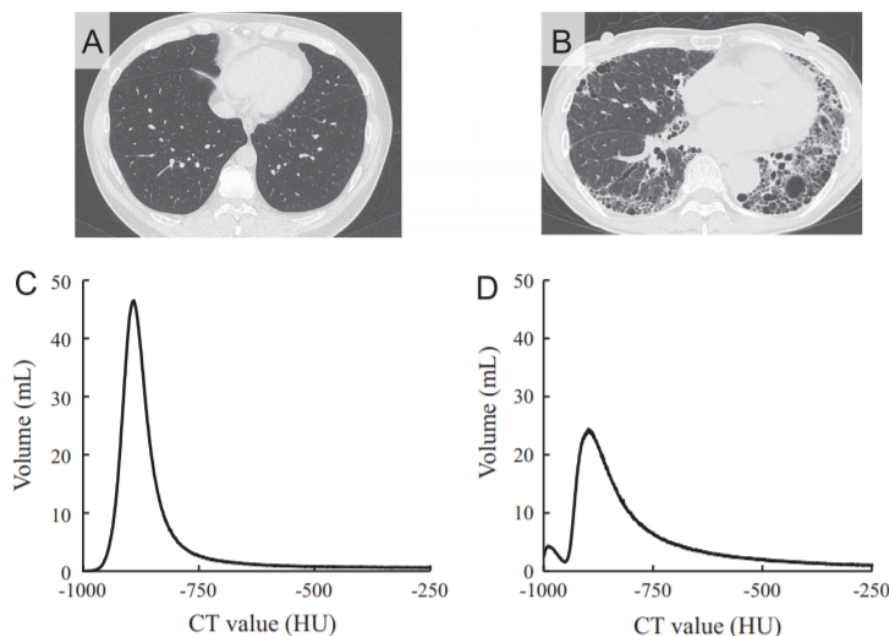


Figure 2.2: Density histogram in patients without and with IPF. (A, B) Computed tomography (CT) images of a healthy participant (A) and a patient with IPF (B)

Increased Radiation Density is observed in the histogram of the patients with IPF compared with that in the healthy participant.

2.2.2 Density mask technique

In Density mask technique, fibrotic changes are quantified by setting threshold to radiation density values to classify different sections of lungs. This helps to differentiate functioning lung regions and regions with fibrotic growth.

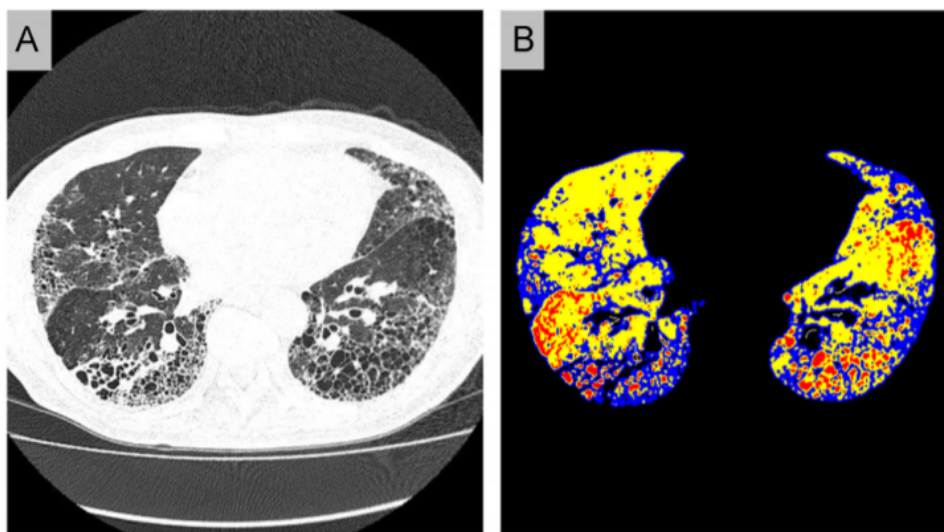


Figure 2.3: (A) Lung Computed tomography image of a person with idiopathic pulmonary fibrosis (IPF). (B) The same CT image, colored based on radiation density in CT image

2.2.3 Texture classification method

This method is used to distinguish between various types of lung regions that reflect pathological findings. Most commonly used texture based Classification method is Computer-Aided Lung Informatics for Pathology Evaluation and Ratings (CALIPER). CALIPER measures Reticular opacity and honey combing in the patients lungs using CT images. This method is mainly used for survival prediction in IPF.

Another method is Gaussian Histogram Normalized Correlation (GHNC). This method compares Local histograms and CT attenuation values along with images, then divides the lungs into different categories. This one is mainly used to find treatment efficiency in patients.

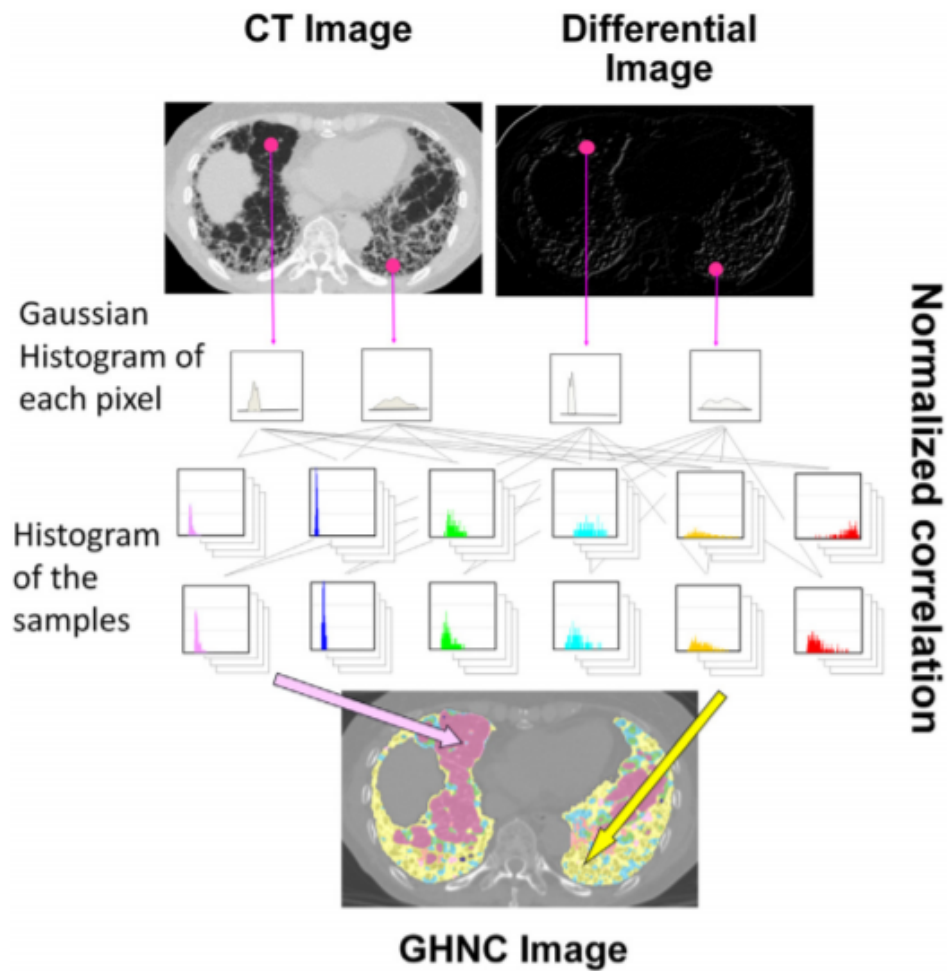


Figure 2.4: The Gaussian Histogram Normalized Correlation system (GHNC). Histograms are collected from each local typical image samples

These texture classification methods are generally computationally heavy, thus requires high processing power.

2.3 CLINICAL COURSE AND PREDICTION OF SURVIVAL IN IDIOPATHIC PULMONARY FIBROSIS (3)

This paper reviews the clinical course, individual predictors of survival, and proposed clinical prediction models in IPF

- Even though there are many clinical and radiographic predictors available, there is no established method of combining these predictors to determine prognosis or define the stage of disease
- The different IPF clinical stages are Subclinical IPF, Slowly Progressive IPF, Rapidly Progressive IPF and Acute Exacerbations of IPF. The progression of IPF varies with different stages of the disease

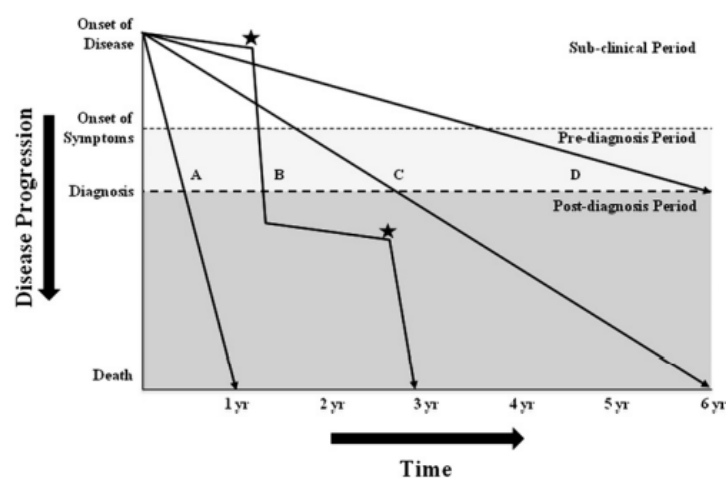


Figure 2.5: Schematic representation of potential clinical courses of IPF

The figure 2.5 shows the rate of progression of IPF in different clinical stages. The rate of decline and progression to death may be rapid (line A), slow (lines C and D), or mixed (curve B), with periods of relative stability interposed with periods of acute decline (star).

- Various clinical predictors of IPF are Age, Smoking status, Sex and FVC. Average survival of patients with different age is shown in table 2.1. IPF is more common in men and mortality rates in women with IPF is increasing rapidly than in men.

Age	Months
Younger than 50	116.4
50-60	62.8
60-70	27.2
Older than 70	14.6

Table 2.1: Average survival of patient with different age

2.4 OBJECTIVES

With this project we intend to develop a Data driven model to predict a patient's severity of decline in lung function based on a Computer Tomography (CT) scan of their lungs, metadata and baseline FVC as input. This project will work towards meeting following goals:

- Reducing the dimensionality and pre-processing the dataset
- Design and train a Machine learning model using features in CT scan and tabular data
- To predict future FVC of the patient based on given CT Scan and tabular data.
- Create a user interface using a web application for potential naive users to use the created model.

Chapter 3

Design and Implementation

3.1 DATASET ANALYSIS

The dataset is taken from the open source imaging consortium database and it is size of more than 23GB. The dataset consists of baseline chest CT scan images of 181 patients and their associated clinical information. A patient has an image acquired at time Week = 0 and has numerous follow up visits over the course of approximately 1-2 years, at which time their FVC is measured

- Patient ID

A unique Id for each patient is used for identify the columns uniquely. This includes name of the patient's Digital Imaging and Communications in Medicine (DICOM) folder

- Weeks

The weeks in the data-set column are the relative number of weeks pre or post the baseline CT. This can be a negative value considering the week in which baseline CT is taken as reference 0.

- FVC

Lung function is assessed based on output from a spirometer, which measures the forced vital capacity. The FVC column is the recorded lung capacity in millilitre. This is the metric that show the declining lung function of the patient.

- Percent

This is a computed field which approximates the patient's FVC as a percent of the typical FVC for a person of similar characteristics.

- Age of patient

IPF is most common in people around age of 65 (4). Age is also determining factor in IPF progression.

- Gender of patient

IPF Progression varies according to gender(5)

- Smoking Status

A recorded variable based on several questions about cigarette smoking. It includes the categories of current smoker, former smoker and never smoked. The data shows that more than 50% of the patients were ex-smokers.

3.2 DATASET PREPROCESSING

Dataset pre-processing is a process of preparing the raw data and making it suitable for a machine learning model. It is the first and crucial step while creating a machine learning model. A real-world data generally contains noises, missing values, and maybe in an unusable format which cannot be directly used for machine learning models. Data preprocessing is required tasks for cleaning the data and making it suitable for a machine learning model which also increases the accuracy and efficiency of a machine learning model.

3.2.1 Crop Boundingbox

Bounding boxes is one of the most popular and recognizable image annotation method used in machine learning and deep learning. A bounding box is an imaginary rectangle that serves as a point of reference for object detection and creates a collision box for that object.

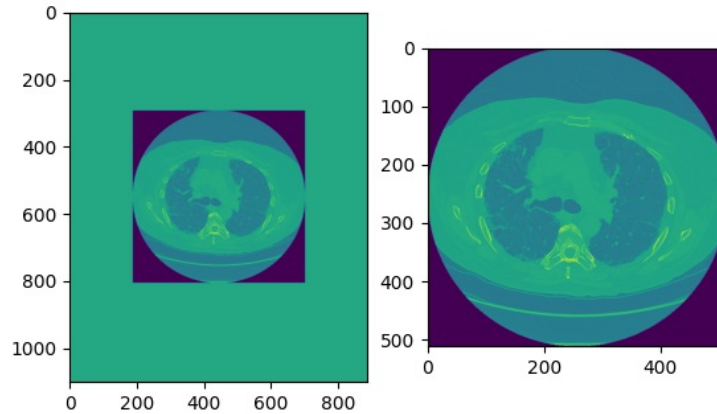


Figure 3.1: Crop boundingbox

Data annotators draw these rectangles over images, outlining the object of interest within each image by defining its X and Y coordinates. This makes it easier for machine learning algorithms to find what they're looking for, determine collision paths, and conserves valuable computing resources. Compared to other image processing methods, this method can reduce costs and increase annotation efficiency.

3.2.2 Converting to Hounsfield Units

Hounsfield scale is a quantitative measure of radio-density and is used to evaluate CT scans. Pixels in an image obtained by CT scanning are displayed in terms of relative radiodensity. The pixel value is displayed according to the mean attenuation of the tissue.

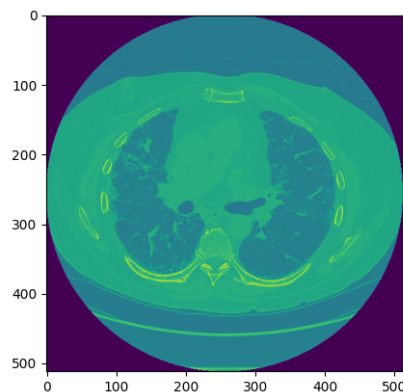


Figure 3.2: Converting to HU

To convert from the normal units found in CT data a linear transformation of the data is applied using the equation

$$HU = \text{PixelValue} * \text{Slope} + \text{Intercept}$$

Slope and Intercept are stored in the DICOM file itself. The tags are generally called the Rescale Slope and Rescale Intercept, and typically have values of 1 and -1024, respectively.

3.2.3 Resizing

Image resizing refers to the changing the scale of images. Scaling comes in handy in many image processing as well as machine learning applications. It helps in reducing the number of pixels from an image and that has several advantages. It can reduce the time of training of a neural network as more is the number of pixels in an image more is the number of input nodes that in turn increases the complexity of the model. Resizing helps in zooming in images.

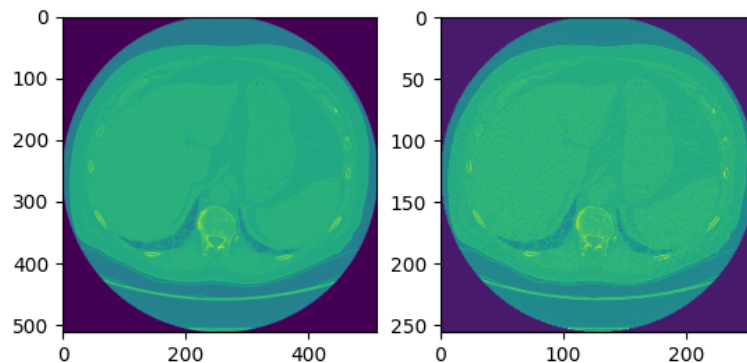


Figure 3.3: Image resizing

Many times we need to resize the image i.e. either shrink it or scale up to meet the size requirements. Images are resized to the dimension 256*256 to make the size of all images are equal.

3.2.4 Clipping

Clipping is used to enhance features within an image. A threshold level is set to determine how the clipping occurs. The values above (or below) the threshold level remain the same while the other values are set equal to the level. The threshold level is applied to every pixels in the image. This is useful for noise reduction when the input signal is known to lie in the range.

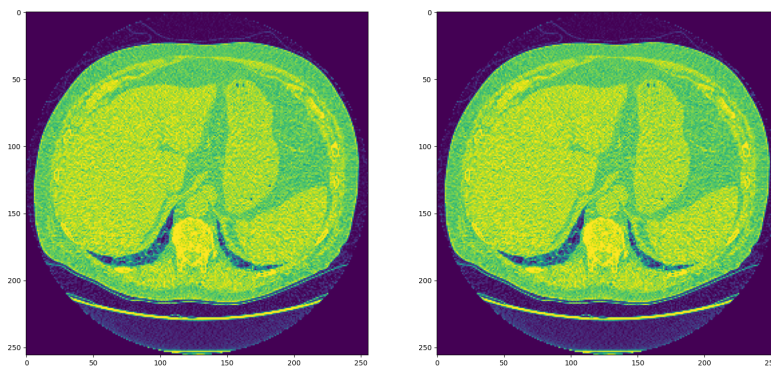


Figure 3.4: Clipping

Intensity values are clipped in the range of -1000 to +500. This function truncates all intensities below the optional value -1000 to all intensities above the optional value to 500. Here clipping is used to remove unwanted features, noise, or extraneous information from an image.

3.2.5 Masking

The concept of a mask is also known as spatial filtering. Mask is a type of filter which performs operation directly on the image. First apply filtering and applying masks is consists of moving the filter mask from point to point in an image. At each point (x,y) of the original image, the response of a filter is calculated by a pre-defined relationship. All the filters values are pre-defined and are a standard.

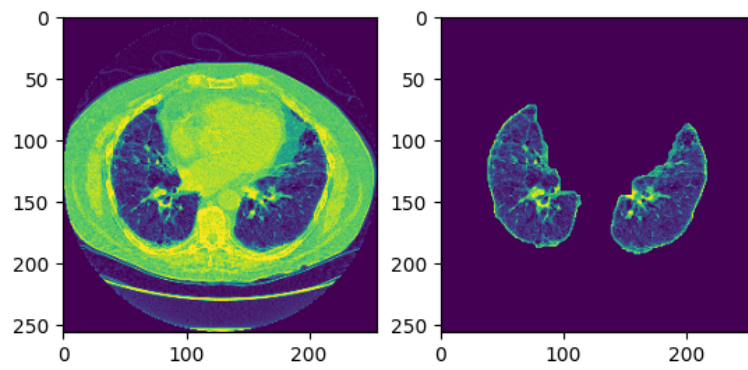


Figure 3.5: Masking

3.3 PRELIMINARY DESIGN

Preliminary design of the project shown in the figure 3.6. The model have two different kinds of data, 3D CT Scan DICOM files and tabular data. It is impossible to design a single neural network model for these two types of data. We divided whole model into two part a Auto encoder part for processing of CT Scan image and a common prediction neural network part. Auto encoder take CT Scan image as input and gives latent features as output. Detailed explanation of Auto encoder is given in section . This latent features will be a array of numerical values so that we can combine with the tabular data.

Prediction part is a simple fully connected neural network with RELU as an activation function. It take combined data of tabular and latent features from the auto encoder and gives the desired out containing FVC value of upcoming weeks.

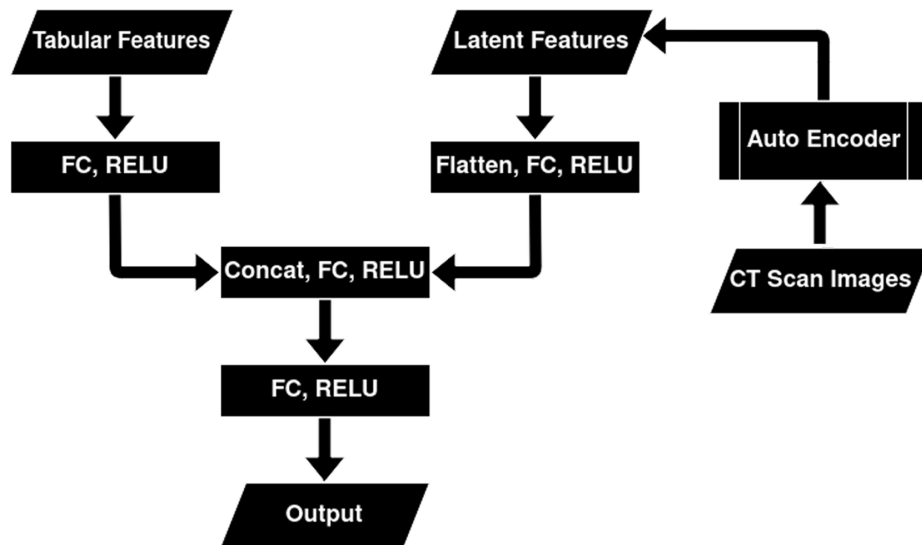


Figure 3.6: Preliminary Design

3.4 AUTO-ENCODER DESIGN

The images after pre-processing is fed into the Auto-Encoder which is made up of an encoder and decoder combined. The Auto-Encoder is designed in such a way that it reduces the dimensions by a huge margin and still be able to re-construct the original image using this reduced dimensions. The design of the proposed Auto-Encoder has been shown in 3.7.

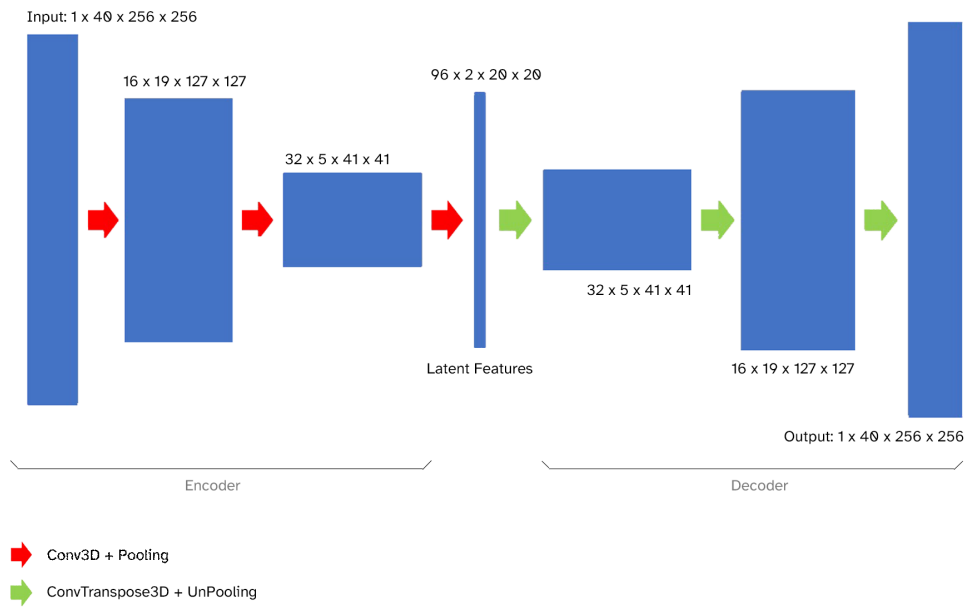


Figure 3.7: Auto-Encoder Design

The pooling layer used for down-sampling the feature map is max pooling. The auto encoder is trained using the training set for 100 epochs and the loss function used is the Mean Square Error (MSE) loss, by comparing the reconstructed samples and the actual samples. The hyperparameters of the autoencoder has been described in table 3.1

Auto-encoder Hyperparameters	
No. of epochs	100
Batch size	32
Learning Rate	0.001
Optimizer	Adam Optimizer

Table 3.1: Autoencoder Hyperparameters

3.5 MULTIPLE QUANTILE REGRESSION DESIGN

The latent features from the Auto-Encoder are fed into the Quantile Model along with the 7 tabular features. The model was trained for 300 epochs. The model takes these data and will give us quantiles. We have designed it as shown in picture 3.8 with 3 quantiles ie the model predicts 3 values.

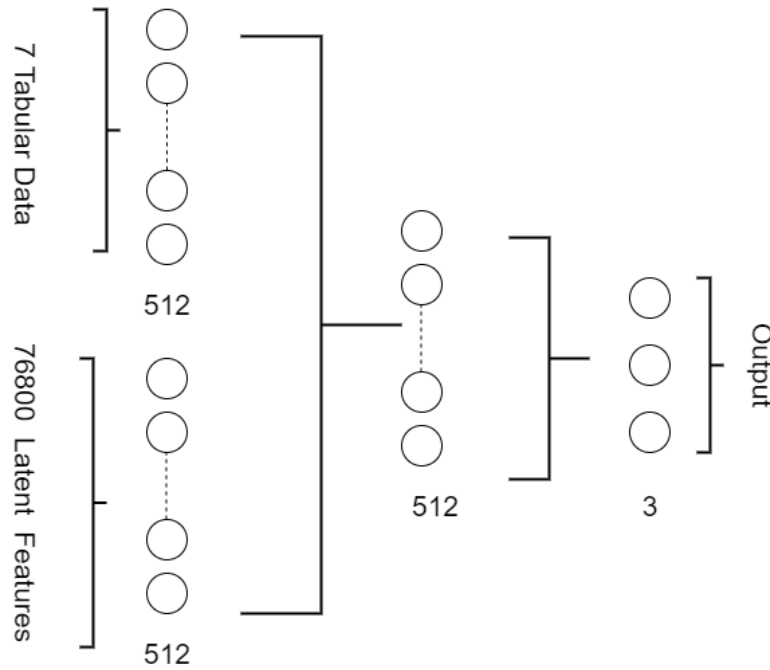


Figure 3.8: Multiple Quantile Regression Model Design

The architecture of the quantile model is shown in Table 3.2.

Layers	Attributes
FullyConnectedLayer1	input=7 tabular features, output=512, ReLu Activation
FullyConnectedLayer2	input=76800 latent features, output=512, ReLu Activation
FullyConnectedLayer3	input=1024(concat 1& 2), output=512, ReLu Activation
FullyConnectedLayer4	input=512, output=3

Table 3.2: Quant Model Architecture

This model uses 3 quantiles of values 0.2 , 0.5 and 0.8 ie it predicts 3 FVC scores with values >20% , >0.5% and >0.8% respectively. These quantiles are used to calculate the confidence interval ie the difference between the 3rd and 1st quantile. The model uses quantile loss function shown below:

$$L(y_i^p, y_i) = \max(q(y_i^p - y_i), (q - 1)(y_i^p - y_i))$$

The hyperparameters of the quantile model is shown in Table 3.3

Quantile Regression Hyperparameters	
No. of epochs	300
Batch size	32
Learning Rate	0.1
Optimizer	Adam Optimizer

Table 3.3: Quantile Regression Hyperparameters

Chapter 4

Results and Discussion

In this chapter, the results obtained from the two models ie AutoEncoder and Multiple Quantile Regression have been specified in detail. The web application developed for the implementation of the models has also been described in the latter part of the chapter. The conclusion of this chapter looks at the existing models present for the task at hand and compares the results obtained from this project with the existing available projects.

4.1 AUTO-ENCODER

The Auto-encoder has been trained for 100 epochs using ADAM optimiser and MSE(Mean Squared Error) function as its loss function. We were able to reduce the training loss to 0.0013851 and validation loss to 0.0014101. The Auto-Encoder was able to reduce the input dimensions of 1x40x256x256 into 96x2x20x20 ie the model was able to reduce the input by approximately 97% and still deliver great results. Fig 4.1 shows one of the test image along with reconstructed image using the output of Auto-Encoder.

Mean Squared Error Function

$$\sqrt{(\frac{1}{n}) \sum_{i=1}^n (y_i - x_i)^2}$$

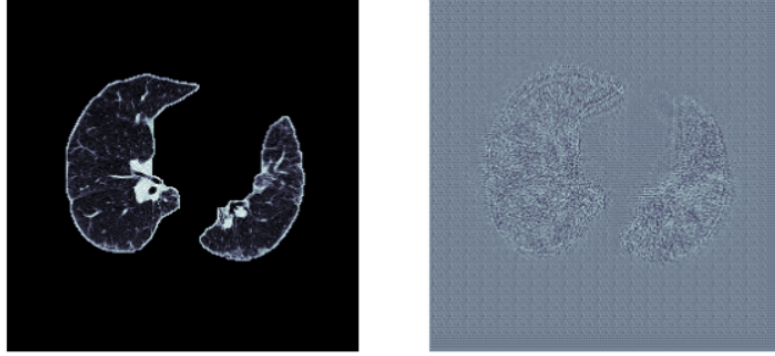


Figure 4.1: Original image and reconstructed image-Auto-Encoder

4.2 MULTIPLE QUANTILE REGRESSION

The Multiple Quantile Regression model has been trained for 300 epochs with cross validation and a loss function of quantile loss. This model was finally valuated based on a loss function called log likelihood. The Likelihood metric score was -6.8466. The model was evaluated using a metric shown below :

$$\sigma_{clipped} = \max(\sigma, 70)$$

$$\Delta = \min(|FVC_{true} - FVC_{predicted}|, 1000)$$

$$metric = -\frac{\sqrt{2}\Delta}{\sigma_{clipped}} - \ln(\sqrt{2}\sigma_{clipped})$$

4.3 WEB APPLICATION

A simple web application was developed for the project which returns the predicted FVC scores as a csv file. The application front-end was developed using simple HTML, CSS and JavaScript. The back-end was implemented using flask framework. The figure 4.2 shows the home page of the web application. The web application lets the user upload the CT scanned images in dcm format along with a csv which has information like age,sex etc. The latent features are extracted and concatenated to the tabular features present in the csv file and the model will predict the future FVC scores along with confidence interval and will deliver a csv file containing this information.

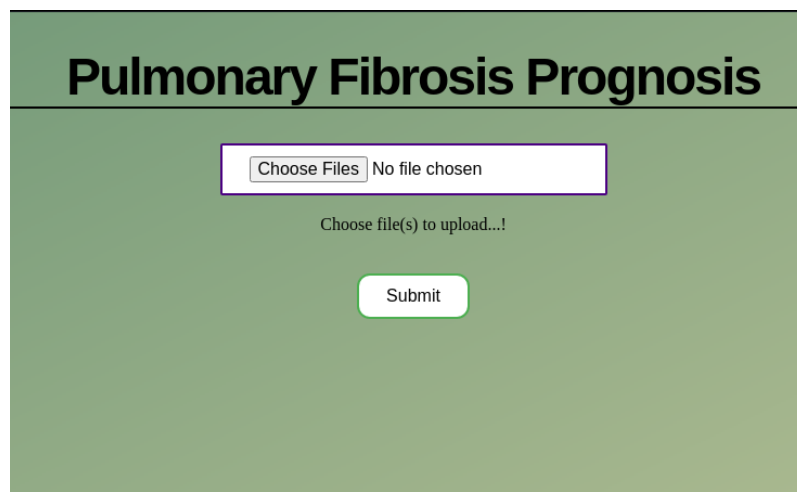


Figure 4.2: Home Page

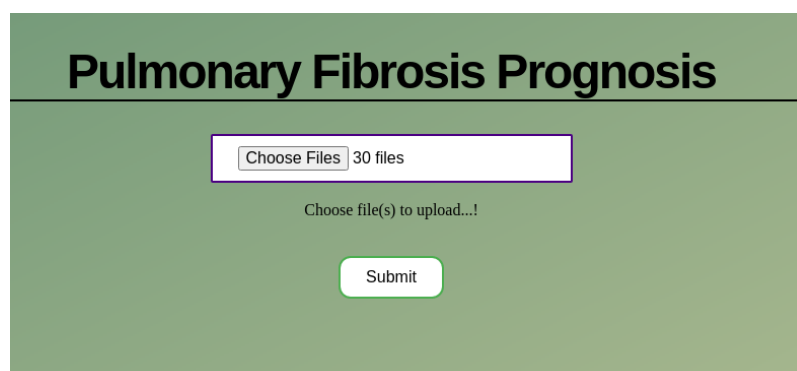


Figure 4.3: After Uploading the files

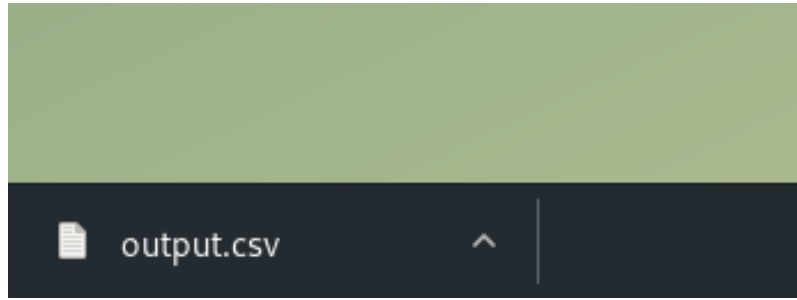


Figure 4.4: Generates a csv

```

output.csv X
run > media > sauravsb99 > BC1C38F91C38AFEE > Dataset > osic-pulmonary-fibrosis-progression > output.csv
1 Patient_Week, FVC, Confidence
2 ID00419637202311204720264 -12, 2569.8706243038177, 140.05747937493425
3 ID00419637202311204720264 -11, 2565.9889229536057, 136.17577802472215
4 ID00419637202311204720264 -10, 2562.1072216033936, 132.29407667451005
5 ID00419637202311204720264 -9, 2558.2255202531815, 128.41237532429795
6 ID00419637202311204720264 -8, 2554.3438189029694, 124.53067397408586
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8 ID00419637202311204720264 -6, 2546.580416202545, 116.76727127366166
9 ID00419637202311204720264 -5, 2542.698714852333, 112.88556992344957
10 ID00419637202311204720264 -4, 2538.817013502121, 109.00386857323747
11 ID00419637202311204720264 -3, 2534.935312151909, 105.12216722302537
12 ID00419637202311204720264 -2, 2531.053610801697, 101.24046587281327
13 ID00419637202311204720264 -1, 2527.1719094514847, 97.35876452260118
14 ID00419637202311204720264 0, 2523.2902081012726, 93.47706317238908
15 ID00419637202311204720264 1, 2519.4085067510605, 89.59536182217698
16 ID00419637202311204720264 2, 2515.5268054008484, 85.71366047196489
17 ID00419637202311204720264 3, 2511.6451040506363, 81.83195912175279
18 ID00419637202311204720264 4, 2507.763402700424, 77.95025777154069
19 ID00419637202311204720264 5, 2503.881701350212, 74.0685564213286
20 ID00419637202311204720264 6, 2500.0, 70.1868550711165

```

Figure 4.5: Outputs FVC and Confidence Interval

4.4 COMPARISON WITH EXISTING WORKS

A comparison with existing models was done, and the result is shown in 4.1

Method	Features	Result	Latency
AutoEncoder + Linear Regression	Light and fast	Accuracy-61%	< 1s
EfficientNet + Multiple Regression	High accuracy	Likelihood score=-6.8305	> 1min
AutoEncoder + Multiple Regression	Light and fairly accurate	Likelihood score=-6.8466	< 10s

Table 4.1: Comparison with Existing models

EfficientNet + Multiple Regression delivers high accuracy with the best metric score but is not suitable for realworld application due to high latency and high resource requirement ie it needs much more GPU power.

AutoEncoder + Linear Regression is the lightest model which requires the least GPU power but is not able to deliver accurate results and the result does not come with a confidence interval like the other two.

AutoEncoder + Multiple Regression is a sweet spot that lies in between these two models ie it is fairly light and delivers accurate results. This is suitable for real life application with acceptable latency. In addition to this, a real time FVC prediction has been achieved using a web application which can be used for medical purposes as shown in figure 4.1.

Chapter 5

Conclusion and Future Scope

CONCLUSION

Pulmonary Fibrosis is a serious disease that affects the lungs and as of now there is no cure. In such a case we need to anticipate the future in order to create a plan that suits a patients characteristics rather than a general medical plan. Our model will make it easier for the medical professionals by predicting the future FVC(Forced Vital Capacity) thereby helping them create a custom plan for patients. Even though there are other models which has a better accuracy than our model, our model is lighter and faster than other models. Our model is apt for the real world use since it can run without high end gpu and cpu.

Our model is designed to give fast and reliable outputs even for large number of dicom files. The model takes both tabular as well as images as input which will extract the latent features from the images which is concatenated with the tabular features to deliver consistent results.

FUTURE SCOPE

Our model is deployed locally using FLASK, the CPU power is capped which can be unlocked by deploying it on a cloud server with a high end GPU such as NVIDIA Titan V which can further reduce the latency.

The GUI presently contains options to upload both the files ie .dcm as well as .csv files as a whole. We can improve by including the lower and upper limit of weeks that need to be predicted thereby eliminating the role of csv file which is more suitable in real life. The GUI can also generate a report using graphical information like graphs or charts which will be better for people using this.

Our model was trained both locally and also using Google Colaboratory which has both GPU and disk constraints. The model was trained for a total of 20epochs which was the maximum achievable value. the model performance can be improved by training the model for at least 100 epochs and can also try varying the learning rate.

Other models which uses other methods such as EfficientNet, Linear Regression can be incorporated into a single model or take averages of the prediction will give us a balanced output.

References

- [1] M. Chen, X. Shi, Y. Zhang, D. Wu and M. Guizani, "Deep Features Learning for Medical Image Analysis with Convolutional Autoencoder Neural Network", in IEEE Transactions on Big Data, doi: 10.1109/TBDATA.2017.2717439.
- [2] Ohkubo H, Nakagawa H, Niimi A. "Computer-based quantitative computed tomography image analysis in idiopathic pulmonary fibrosis: A mini review". *Respir Investig*. 2018 Jan;56(1):5-13. doi: 10.1016/j.resinv.2017.10.003. Epub 2017 Dec 7. PMID: 29325682.
- [3] Ley B, Collard HR, King TE Jr. "Clinical course and prediction of survival in idiopathic pulmonary fibrosis". *Am J Respir Crit Care Med*. 2011 Feb 15;183(4):431-40. doi:10.1164/rccm.201006-0894CI. Epub 2010 Oct 8. PMID: 20935110.
- [4] Collard HR. The age of idiopathic pulmonary fibrosis. *Am J Respir Crit Care Med* 2010;181:771–772.
- [5] Kärkkäinen, Miia, et al. "Comparison of disease progression subgroups in idiopathic pulmonary fibrosis." *BMC pulmonary medicine* 19.1 (2019): 1-9.